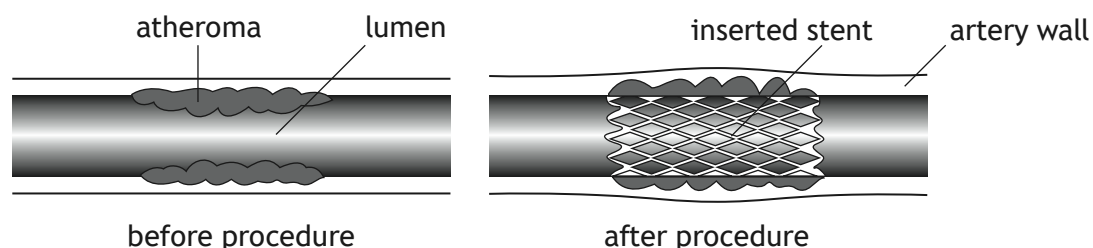


8. When an atheroma develops in an artery it can be treated by inserting a mesh tube called a stent.

The diagrams show an artery before and after a stent has been inserted.



- (a) (i) Describe the formation of an atheroma within an artery.

2

- (ii) Use the diagrams to suggest how the insertion of a stent leads to increased blood flow in an artery.

1

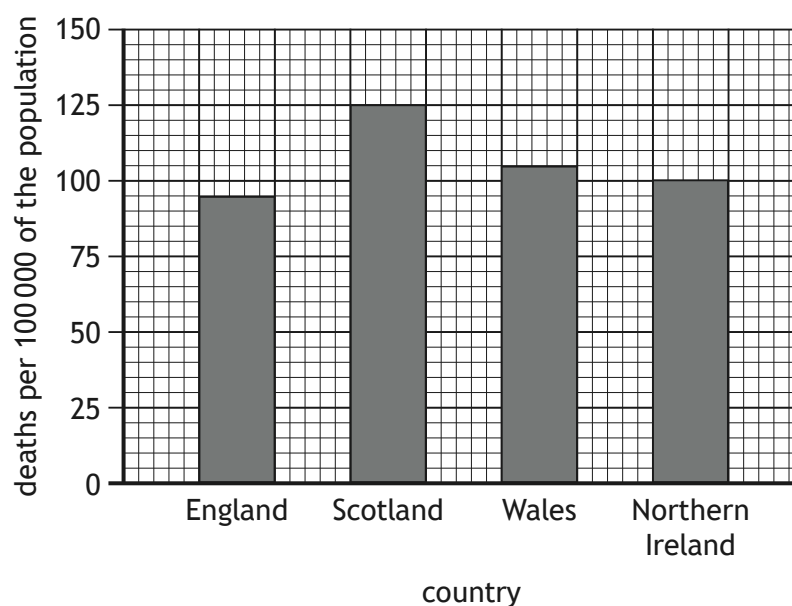
- (b) A ruptured atheroma in an artery may lead to thrombosis.

Describe how thrombosis can cause a myocardial infarction (heart attack).

2

8. (continued)

(c) The graph shows deaths from coronary heart disease in four countries.



- (i) Using data from the graph, the risk of an individual dying from coronary heart disease in Northern Ireland was calculated to be 1 in 1000.

Use data from the graph to calculate the risk of an individual dying from coronary heart disease in Scotland.

1

Space for calculation

1 in _____

- (ii) Explain why deaths from coronary heart disease are expressed as per 100 000 of the population.

1

[Turn over

